

SOCIOL 723: Practice Midterm, With Answers

For practice before the in-class midterm on Thursday, October 15

Instructions. This practice exam has the same format, length, and level as the midterm. The midterm is open book and open note, timed (150 minutes), and asks for no code. It covers Weeks 1–7 and draws on the frames marked with a blue star in the slides. Where a question asks for sentences, write them as you would in a paper. Show your arithmetic: a right number with no work earns partial credit only.

Part	Format	Weeks	Points
1	Five yes-or-no questions	1–7	10
2	Ten single-choice questions	1–7	30
3	Three longer questions	1–2, 3–4, 5 and 7	60
Bonus	One derivation	5–6	+5

Part 1: Yes or No

Circle YES or NO. Two points each.

1. In your sample, the OLS residuals are uncorrelated with X . This is evidence that $E[\epsilon_i | X_i] = 0$. YES NO

Answer. NO. The normal equations force $\sum_i X_i \hat{\epsilon}_i = 0$ in every sample, for every dataset. A zero correlation between residuals and X is algebra, not evidence.

2. Heteroskedasticity makes the OLS coefficients biased. YES NO

Answer. NO. Under exogeneity OLS stays unbiased. Only the classical standard-error formula is wrong; the robust (sandwich) formula fixes it.

3. The likelihood $\mathcal{L}(\theta; y)$ is a probability distribution over θ . YES NO

Answer. NO. It is the density read backwards: data fixed, θ varying. It does not integrate to one over θ .

4. If treatment is assigned by a coin flip, the difference in mean outcomes between the treated and the untreated estimates the ATE. YES NO

Answer. YES. Randomization gives $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$, so the selection-bias term $E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$ is zero and $ATE = ATT$.

5. With a binary instrument and no defiers, the first stage $E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]$ equals the share of compliers. YES NO

Answer. YES. The first stage is $\Pr(C) - \Pr(\text{defiers})$, and monotonicity sets the second term to zero.

Part 2: Single Choice

Circle one letter. Three points each.

1. A researcher regresses children's test scores on neighborhood poverty and omits parental income. Parental income raises test scores, and it is lower in poorer neighborhoods. The short regression's poverty coefficient is
 - (A) biased downward: poverty looks more harmful than it is.
 - (B) biased upward: poverty looks less harmful than it is.
 - (C) unbiased, because parental income only affects the outcome.
 - (D) impossible to sign without the data.

Answer. (A). Bias = $\beta_2 \delta_1$ with $\beta_2 > 0$ (income raises scores) and $\delta_1 < 0$ (income is lower where poverty is higher): the bias is negative.

2. Students sit in 40 schools, and the treatment was assigned by school. A pairs bootstrap that resamples *students* gives a standard error of 0.8. The right bootstrap
 - (A) resamples whole schools with replacement, each bringing all its students.
 - (B) resamples students, but with $B = 5,000$ instead of 1,000.
 - (C) resamples students within each school.
 - (D) is not needed: the pairs bootstrap is already heteroskedasticity-robust.

Answer. (A). Resample the units the assignment lottery ran over. Resampling students copies the too-small standard error of the naive formula.

3. Five independent Poisson counts: 2, 5, 1, 4, 3. The MLE $\hat{\theta}$ and its standard error are
 - (A) 3 and 0.77
 - (B) 3 and 0.60
 - (C) 3 and 1.73
 - (D) 2.5 and 0.71

Answer. (A). $\hat{\theta} = \bar{y} = 15/5 = 3$. Observed information $-\ell''(\hat{\theta}) = \sum_i y_i / \hat{\theta}^2 = 15/9 = 1.67$, so $SE = 1/\sqrt{1.67} = 0.77$; equivalently $\sqrt{\hat{\theta}/n} = \sqrt{3/5}$.

4. A logit is fitted with and without a block of three dummies, on the same 2,000 rows. The log-likelihoods are $-1,199.1$ (with) and $-1,204.6$ (without). The χ^2_3 5% cut-off is 7.81 and $\log 2000 = 7.6$. Which is right?

- (A) $LR = 11.0$: reject at 5%; BIC prefers the model without the block.
- (B) $LR = 5.5$: do not reject at 5%.
- (C) $LR = 11.0$: reject at 5%; BIC prefers the model with the block.
- (D) $LR = -11.0$: the smaller model fits better.

Answer. (A). $LR = 2[-1199.1 - (-1204.6)] = 11.0 > 7.81$. BIC charges $q \log n = 3 \times 7.6 = 22.8$ for the block, more than the 11.0 it buys: $BIC_{\text{small}} - BIC_{\text{big}} = 11.0 - 22.8 < 0$.

5. In a logit of voting (0/1), the coefficient on holding a college degree is 0.693, so $e^{0.693} = 2$. Which sentence is correct?
- (A) Graduates are twice as likely to vote as non-graduates.
 - (B) A degree raises the probability of voting by 0.693.
 - (C) A degree doubles the odds of voting, other covariates held fixed.
 - (D) A degree raises the probability of voting by 50 percentage points.

Answer. (C). e^β multiplies the odds. It is not a ratio of probabilities, and the change in probability depends on where a person starts on the S-curve.

6. A Poisson regression of the number of arrests shows strong overdispersion. What follows?
- (A) The coefficients are badly biased and must be discarded.
 - (B) The coefficients are about right, but the Poisson standard errors are too small.
 - (C) The standard errors are too large, so the tests are conservative.
 - (D) Nothing: overdispersion only matters for prediction.

Answer. (B). The mean is still right, so $\hat{\beta}$ is fine; the Poisson's variance = mean claim is wrong, so its SEs are too small. Use robust SEs or a negative binomial.

7. A random-intercept model of math scores for students in schools reports $\hat{\tau}^2 = 4$ (between schools) and $\hat{\sigma}^2 = 36$ (within schools). The intraclass correlation is
- (A) 0.10
 - (B) 0.11
 - (C) 0.33
 - (D) 0.90

Answer. (A). $ICC = \tau^2 / (\tau^2 + \sigma^2) = 4/40 = 0.10$: 10% of the unexplained variation is between schools.

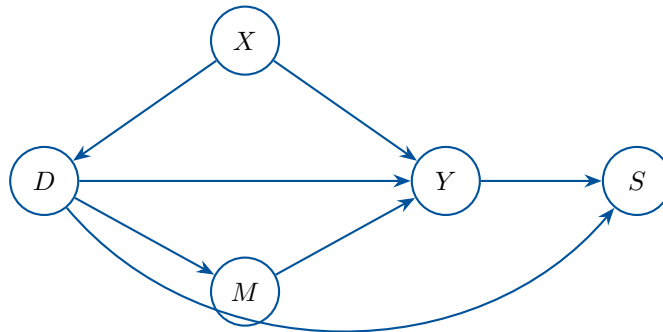
8. Four units:

Unit	D_i	$Y_i(0)$	$Y_i(1)$
1	1	3	9
2	1	5	9
3	0	4	6
4	0	2	2

- (A) ATE = 3, ATT = 5, naive comparison = 6
- (B) ATE = 5, ATT = 3, naive comparison = 6
- (C) ATE = 3, ATT = 5, naive comparison = 5
- (D) ATE = 4, ATT = 5, naive comparison = 6

Answer. (A). $\tau_i = 6, 4, 2, 0$. ATE = $12/4 = 3$; ATT = $(6 + 4)/2 = 5$. Observed $Y_i = 9, 9, 4, 2$, so naive = $9 - 3 = 6 = \text{ATT} + \text{selection bias} = 5 + (4 - 3)$.

9. In the DAG below, D is job training, Y is earnings two years later, X is earnings before training, M is hours worked after training, and S is whether the person is still in the survey at follow-up. To estimate the *total* effect of D on Y , condition on



- (A) X only.
- (B) X and M .
- (C) X and S (that is, analyze only people still in the survey).
- (D) X , M , and S .

Answer. (A). X is a confounder (the back door $D \leftarrow X \rightarrow Y$). M is a mediator: conditioning removes part of the effect. S is a collider ($D \rightarrow S \leftarrow Y$): conditioning, including by keeping only respondents still in the survey, opens a spurious path.

10. In a cell where the propensity score is $e(x) = 0.8$, weighting for the ATT gives each *untreated* unit the weight
- (A) 4
 - (B) 1.25
 - (C) 5
 - (D) 0.25

Answer. (A). ATT weights: treated 1, untreated $e(x)/(1 - e(x)) = 0.8/0.2 = 4$, the odds of treatment. The untreated are reweighted to look like the treated, of whom there are four per untreated unit in this cell.

Part 3: Longer Questions

Twenty points each. Every number below comes from the course's GSS extract or from Lab 7's data.

Question 1: Schooling, Parental Education, and Earnings

The GSS extract has 3,509 full-time workers aged 25–64 (GSS 2010–22). The outcome is log annual earnings. Three OLS regressions:

	(1) log earnings	(2) log earnings	(3) parents' education
years of schooling	0.1187	0.1099	0.5703
parents' education (years)		0.0154	
intercept	8.2307	8.1546	4.9405
R^2	0.129	0.132	

Standard errors for the schooling coefficient in model (2), three ways: classical 0.0058; HC3 0.0064; clustered on the GSS primary sampling unit (839 clusters, median size four) 0.0065.

- (a) Interpret the schooling coefficient in model (1) in one sentence a reader of a sociology journal would understand.

Answer. Among these workers, each additional year of schooling is associated with about 0.119 log points, roughly 12%, higher annual earnings. It is a description (the best linear approximation to the CEF), not yet a causal claim.

- (b) Use the omitted-variable-bias formula to recover the coefficient in model (1) from models (2) and (3). Show the arithmetic, and explain in words why adding parents' education lowers the schooling coefficient.

Answer. Short = long + (effect of omitted) $\times$ (regression of omitted on included): $0.1099 + 0.0154 \times 0.5703 = 0.1099 + 0.0088 = 0.1187$. Parents' education raises earnings ($0.0154 > 0$) and moves with own schooling ($0.5703 > 0$), so model (1) credits schooling with part of the family-background advantage; the bias is positive.

- (c) Is 0.1099 the causal effect of a year of schooling? Name one variable still omitted, sign the bias it creates, and say whether a higher R^2 would settle the matter.

Answer. Not without an argument that nothing else confounds schooling and earnings. Ability is the standard omitted variable: it raises earnings and rises with schooling, so the coefficient is still overstated. R^2 does not help: it answers how well the line fits (Week 1's Q1), not whether the compared people are comparable (Q2).

- (d) Using the HC3 standard error, compute the t -statistic and a 95% confidence interval for the schooling coefficient in model (2). Explain why the HC3 and classical standard errors differ while the coefficient is the same.

Answer. $t = 0.1099/0.0064 = 17.2$; interval $0.1099 \pm 1.96 \times 0.0064 = [0.097, 0.122]$. The coefficient comes from the same normal equations either way. The classical formula pools one s^2 ; HC3 lets each observation carry its own squared residual (inflated by its leverage), so it is right when the spread of earnings differs across schooling levels. Heteroskedasticity changes the uncertainty, not the estimate.

- (e) The GSS samples areas first and then people within them. Which of the three standard errors would you report, and why are the HC3 and clustered numbers so close here?

Answer. The clustered one: cluster at the level of the sampling lottery, the primary sampling unit. Here the clusters are tiny (median four people), so little within-cluster correlation can accumulate and the clustered SE barely moves (0.0065 vs 0.0064). Say so rather than choosing the smaller number.

- (f) A colleague suggests adding occupational prestige to model (2) “to be safe.” Do you agree? Answer with a reason that refers to what the coefficient would then estimate.

Answer. No. Occupation is a consequence of schooling on the path to earnings, a mediator. Holding it fixed removes the part of the schooling effect that works through better jobs, so the coefficient would no longer be the total effect. More controls are not more conservative.

Question 2: A Logit of Ever Having Married

Same 3,509 workers; 73% have ever married. A logit of ever married (0/1):

	$\hat{\beta}$	SE	z	$e^{\hat{\beta}}$
intercept	-2.2286	0.2791	-7.98	
years of schooling	-0.0147	0.0149	-0.99	0.985
female	0.0406	0.0836	0.49	1.041
Black	-0.9636	0.1066	-9.04	0.382
age (years)	0.0864	0.0043	19.91	1.090
log-likelihood	-1,769.5			

Average marginal effects: age 0.0144 per year; Black -0.178.

- (a) Write the model twice: once in log-odds form and once in probability form.

Answer. $\log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 \text{school}_i + \beta_2 \text{female}_i + \beta_3 \text{Black}_i + \beta_4 \text{age}_i = X_i' \beta$, and

$$p_i = \frac{e^{X_i' \beta}}{1 + e^{X_i' \beta}}.$$

- (b) Read the Black coefficient on all three scales: log-odds, odds, and probability. Say what is wrong with the sentence “Black workers are 62% less likely to have married.”

Answer. Log-odds: 0.96 lower, other covariates fixed. Odds: multiplied by $e^{-0.964} = 0.38$. Probability: on average 17.8 percentage points lower (the AME). The sentence reads the odds ratio as a ratio of probabilities; 0.38 multiplies the odds, and the probability change depends on where a person sits on the S-curve.

- (c) Compute, by hand, the predicted probability of having married for a woman with 16 years of schooling who is not Black, at age 30 and at age 40.

Answer. Age 30: $X' \hat{\beta} = -2.2286 - 0.0147(16) + 0.0406 + 0.0864(30) = 0.169$; odds $e^{0.169} = 1.18$; $\hat{p} = 1.18/2.18 = 0.54$. Age 40: add $10 \times 0.0864 = 0.864$, $X' \hat{\beta} = 1.033$, odds 2.81, $\hat{p} = 0.74$.

- (d) For this woman, ten years of age raise the probability by about 20 points. Ten times the AME of age is only 14 points. Explain the difference.

Answer. The slope of the probability is $\hat{p}_i(1 - \hat{p}_i)\hat{\beta}$: steepest at $\hat{p} = 0.5$. This woman starts at 0.54, the steep part of the S-curve. The AME averages the slope over all 3,509 people, and with 73% married most of them sit higher, where the curve is flatter.

- (e) Adding eight census-region dummies raises the log-likelihood to $-1,756.6$. Carry out the likelihood-ratio test (the χ^2_8 5% cut-off is 15.5). Then compute $BIC_{\text{without}} - BIC_{\text{with}}$ using $\log n = 8.16$, and explain why the test and BIC disagree. If region were a confounder of a coefficient you care about, would you drop it?

Answer. $LR = 2[-1756.6 - (-1769.5)] = 25.9 > 15.5$, $p = 0.001$: region matters. BIC: $LR - q \log n = 25.9 - 8 \times 8.16 = -39.4$, so BIC prefers the model without region. Eight parameters bought 25.9 units of fit, about 3.2 each: too much for chance, too little for BIC's price of 8.16 per parameter. A confounder stays regardless: selection criteria choose descriptions, not identification.

Question 3: College Proximity as an Instrument (Card 1995)

Lab 7's data: 3,010 men from the NLS Young Men, first interviewed in 1966. D = years of schooling; Y = log hourly wage in 1976; $Z = 1$ if the man grew up in a county with a four-year college (68% did). All four regressions control for experience, experience squared, Black, metropolitan residence, and South. HC1 standard errors in parentheses:

	OLS of Y on D	first stage, D on Z	reduced form, Y on Z	IV
estimate	0.074	0.337	0.045	0.132
(se)	(0.004)	(0.081)	(0.016)	(0.049)

Robust first-stage $F = 17.5$. (Unrounded: reduced form 0.0446, first stage 0.3373.)

- (a) Draw a DAG with Z , D , Y , unmeasured ability U , and local labor-market conditions L . Mark the arrows that must be *absent* for Z to be a valid instrument.

Answer. Arrows present: $Z \rightarrow D \rightarrow Y$, $U \rightarrow D$, $U \rightarrow Y$; $L \rightarrow Y$ (and possibly $L \rightarrow D$). Must be absent: $Z \rightarrow Y$ directly (exclusion), and any common cause of Z and Y , such as $L \rightarrow Z$ with $L \rightarrow Y$, or $U \rightarrow Z$ (independence). The controls (metropolitan, South) are there to block the $Z \leftarrow L \rightarrow Y$ path.

- (b) Show that the IV estimate is the ratio of two numbers in the table, and interpret it in one sentence.

Answer. $\hat{\beta}_{IV} = \text{reduced form} / \text{first stage} = 0.0446 / 0.3373 = 0.132$. Growing up near a college raises wages by 4.5% and schooling by a third of a year; scaled up, a year of schooling raises wages by about 13% for the men whose schooling proximity changed.

- (c) State the three IV conditions for this instrument. For each, say whether it can be tested, and give one threat specific to college proximity.

Answer. Relevance: proximity moves schooling. Testable; passes (0.337, $F = 17.5$). **Independence:** proximity is as good as random given the controls. Not testable; threat: families who live near colleges differ (richer, more educated parents). **Exclusion:** proximity affects wages only through schooling. Not testable; threat: counties with colleges have better local labor markets that raise wages directly.

- (d) Ability bias predicts that OLS overstates the return to schooling, yet IV (0.132) is almost twice OLS (0.074). Give two explanations consistent with valid instruments.

Answer. (i) **Measurement error** in reported schooling pulls OLS toward zero; IV is not affected by classical error in D . (ii) **LATE**: IV estimates the return for compliers, men whose schooling was moved by having a college nearby, typically from poorer families facing cost barriers; their return can be higher than the average return OLS summarizes. (A violated exclusion or weak-instrument bias would also push IV up, but those are failures of the design, not explanations under valid instruments.)

- (e) Treat schooling as “attended college or not.” Describe compliers, always-takers, and never-takers in this setting, and state monotonicity in words. Is the IV estimate the right number for a policy that opens new colleges in counties without one?

Answer. Compliers attend college only if one is nearby; always-takers attend wherever they grow up; never-takers do not attend either way. Monotonicity: no one attends *because* there is no college nearby. For opening new colleges the LATE is close to the right number: the men such a policy would move are exactly those whose attendance responds to proximity. It is not the effect for always-takers or for everyone.

- (f) The first-stage F of 17.5 clears the “ $F > 10$ ” rule. Should you trust the conventional t -test of the IV estimate ($t = 2.72$)? What would you report?

Answer. Not fully. $F > 10$ was derived to keep bias small; for a t -test with correct size, Lee et al. (2022) show F must exceed about 104. At $F = 17.5$ the honest critical value is well above 1.96, so report a weak-instrument-robust interval (Anderson–Rubin, or the tF adjustment), together with the reduced form (0.045, $t = 2.72$), which needs no strength assumption, and the first stage and OLS alongside.

Bonus: Where the Naive Comparison Goes Wrong

Five points. Use only the definitions of the ATE, ATT, ATU, and $\pi = \Pr(D_i = 1)$, the switching equation, and the law of iterated expectations.

(a) Prove that $\text{ATE} = \pi \text{ATT} + (1 - \pi) \text{ATU}$.

(b) Prove that

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= \text{ATE} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline selection}} \\ &\quad + \underbrace{(1 - \pi)(\text{ATT} - \text{ATU})}_{\text{selection on the effect}}. \end{aligned}$$

(c) Check the identity on the four units of Part 2, Question 8.

(d) Show that both selection terms are zero when treatment is randomly assigned.

Answer. (a) Write $\tau_i = Y_i(1) - Y_i(0)$. By iterated expectations over the two values of D_i ,

$$\begin{aligned} \text{ATE} &= E[\tau_i] = \Pr(D_i = 1) E[\tau_i \mid D_i = 1] + \Pr(D_i = 0) E[\tau_i \mid D_i = 0] && \text{iterated exp.} \\ &= \pi \text{ATT} + (1 - \pi) \text{ATU}. && \text{definitions} \end{aligned}$$

(b) By the switching equation, $Y_i = Y_i(1)$ when $D_i = 1$ and $Y_i = Y_i(0)$ when $D_i = 0$. Add and subtract $E[Y_i(0) \mid D_i = 1]$:

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] \\ &= \text{ATT} + E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]. \end{aligned}$$

From (a), $\text{ATT} - \text{ATE} = \text{ATT} - \pi \text{ATT} - (1 - \pi) \text{ATU} = (1 - \pi)(\text{ATT} - \text{ATU})$, so $\text{ATT} = \text{ATE} + (1 - \pi)(\text{ATT} - \text{ATU})$. Substituting this for ATT in the second line gives the identity.

(c) Naive = 6; ATE = 3; baseline selection = $\frac{1}{2}(3 + 5) - \frac{1}{2}(4 + 2) = 1$; $\pi = 0.5$, $\text{ATU} = (2 + 0)/2 = 1$, so $(1 - \pi)(\text{ATT} - \text{ATU}) = 0.5 \times (5 - 1) = 2$. $3 + 1 + 2 = 6$. ✓ (d) Randomization gives $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$. Then $E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0]$, so baseline selection is zero; and $E[\tau_i \mid D_i = 1] = E[\tau_i \mid D_i = 0]$, so $\text{ATT} = \text{ATU}$ and the second term is zero.